

CUSTOMIZABLE SUBSCRIPTION BOX SCHEDULER

Software Architecture & Design Specification
Problem Statement #35

Student Name: SAIMA KAPOOR
SRN: PES1UG24CS403
Course: Software Engineering
Lab 3

1. Architectural Pattern

Architectural Pattern: Layered Architecture

The Customizable Subscription Box Scheduler uses a Layered Architecture.

The system is organized into three logical layers:

1. Presentation Layer
2. Business/Application Layer
3. Data Layer

The Presentation Layer provides interfaces for the Subscriber and Fulfillment Lead.

The Business/Application Layer contains the core business logic for subscription management, box customization, billing-cycle and deadline validation, fulfillment management, and fulfillment manifest generation.

The Data Layer stores subscriber, subscription, box customization, billing-cycle, and fulfillment information.

The layers communicate through defined interfaces, with the Presentation Layer accessing the Business/Application Layer and the Business/Application Layer accessing the Data Layer.

Why this architecture?

Architectural Justification

Layered Architecture was selected because the system has a clear separation between user interaction, business rules, and persistent data.

The architecture separates presentation logic from business logic, making the system easier to maintain and modify. For example, the 48-hour customization deadline can be implemented within the Billing and Deadline Service without placing this business rule inside the user interface. The architecture also supports modularity. Subscription management, box customization, billing validation, and fulfillment functionality can be developed and tested as separate components. The Fulfillment Manifest Generator is separated from the other fulfillment functions because the system has a performance requirement for generating shipping labels for a large number of monthly boxes. The separation of components also makes it easier to trace system functionality back to the functional and non-functional requirements defined in Lab 1.

2. Component Description and Requirement Traceability

Component	Responsibility	Related Requirements
Subscriber Interface	Provides subscriber access to subscription and box customization functions.	FR-001, FR-002, FR-003
Subscription Management Service	Manages subscription status, pausing, skipping and resuming billing cycles.	FR-003
Box Customization Service	Manages product selection and item swapping within a subscription box.	FR-001, FR-002
Billing & Deadline Service	Validates whether customization or subscription changes are permitted before the applicable deadline.	FR-001, FR-002, FR-003

Fulfillment Lead Interface	Allows the Fulfillment Lead to access fulfillment information.	FR-004
Fulfillment Management Service	Manages fulfillment information and keeps the fulfillment data synchronized with subscriber selections.	FR-004, FR-005
Fulfillment Manifest Generator	Generates the fulfillment manifest and shipping information for upcoming boxes.	FR-004, NFR-001
CSBS Database	Stores subscriber, subscription, product selection, billing-cycle and fulfillment information.	FR-001–FR-005, NFR-002

3. System Components

3.1 Subscriber Interface

The Subscriber Interface provides the user interface through which subscribers interact with the system.

The interface allows subscribers to:

- View their subscription
- Customize their monthly subscription box
- Swap eligible box items
- Pause their subscription
- Skip a billing cycle
- Resume their subscription

3.2 Fulfillment Lead Interface

The Fulfillment Lead Interface provides access to fulfillment-related functionality.

The interface allows the Fulfillment Lead to:

- View fulfillment information
- View upcoming subscription boxes
- Access fulfillment manifests
- Access information required for preparing shipments

3.3 Subscription Management Service

The Subscription Management Service manages subscriber subscription information and subscription lifecycle operations.

Its responsibilities include:

- Viewing subscription information
- Pausing a subscription
- Skipping a billing cycle
- Resuming a subscription
- Updating subscription status
- Maintaining billing-cycle information

3.4 Box Customization Service

The Box Customization Service manages the contents of a subscriber's monthly subscription box.

Its responsibilities include:

- Customizing the monthly box
- Selecting eligible products
- Swapping eligible products
- Validating product selections
- Saving updated box selections

3.5 Billing and Deadline Service

The Billing and Deadline Service manages billing-cycle timing and validates whether subscriber changes are allowed.

Its responsibilities include:

- Checking the billing renewal date
- Validating the applicable customization deadline
- Validating pause and skip requests
- Preventing changes after the applicable deadline
- Providing billing-cycle information to other components

3.6 Fulfillment Management Service

The Fulfillment Management Service manages information required to prepare subscription boxes for fulfillment.

Its responsibilities include:

- Retrieving upcoming subscription boxes
- Retrieving subscriber product selections
- Maintaining fulfillment information
- Providing fulfillment data to the manifest generator
- Keeping fulfillment information synchronized with subscriber changes

3.7 Fulfillment Manifest Generator

The Fulfillment Manifest Generator generates fulfillment information required for shipment preparation.

Its responsibilities include:

- Collecting finalized subscription-box information
- Generating the fulfillment manifest
- Preparing shipping information
- Exporting shipping labels
- Processing a large number of monthly subscription boxes efficiently

3.8 CSBS Database

The CSBS Database provides persistent storage for the system.

The database stores information such as:

- Subscriber information
- Subscription information
- Product information
- Box selections
- Customization information
- Billing-cycle information
- Fulfillment information
- Manifest information

4. Component Interaction

The Subscriber Interface communicates with the Subscription Management Service, Box Customization Service, and Billing and Deadline Service to perform subscriber operations. The Subscription Management Service manages subscription status, pause operations, skip operations, resume operations, and billing-cycle information. The Box Customization Service manages product selections and item swaps within a subscriber's monthly box. The Billing and Deadline Service validates whether requested subscription or customization changes are permitted according to the applicable deadline. The Fulfillment Lead Interface communicates with the Fulfillment Management Service to access fulfillment information. The Fulfillment Management Service retrieves the latest subscription and customization information and provides the required information to the Fulfillment Manifest Generator. The CSBS Database provides

persistent storage for subscription, customization, billing-cycle, and fulfillment information.

5. Component Interaction Flow

The major interaction flow of the system is:

Subscriber → Subscriber Interface → Subscription Management Service / Box Customization Service / Billing and Deadline Service → CSBS Database

For fulfillment operations:

Fulfillment Lead → Fulfillment Lead Interface → Fulfillment Management Service → Fulfillment Manifest Generator → CSBS Database

The overall flow ensures that subscriber changes are stored and that the latest valid subscription information is available to the fulfillment process.

6. Requirement Traceability

Component	Main Responsibility	Related Requirements
Subscriber Interface	Provides subscriber access to subscription and customization functions	FR-001, FR-002, FR-003
Subscription Management Service	Manages subscription status and billing-cycle operations	FR-003
Box Customization Service	Manages product selection and item swapping	FR-001, FR-002
Billing and Deadline Service	Validates subscription and customization deadlines	FR-001, FR-002, FR-003
Fulfillment Lead Interface	Provides access to fulfillment functionality	FR-004

Fulfillment Management Service	Manages fulfillment information and synchronization	FR-004, FR-005
Fulfillment Manifest Generator	Generates fulfillment manifests and shipping information	FR-004, NFR-001
CSBS Database	Stores system and fulfillment data	FR-001 to FR-005, NFR-002

7. Security Architecture

The system shall restrict subscriber functionality to authenticated subscribers. A subscriber shall only be able to view and modify their own subscription information. Fulfillment functionality shall be restricted to authorized Fulfillment Lead users. Role-based authorization shall be used to control access to subscriber and fulfillment functionality. The system shall validate subscription customization, pause, and skip operations before updating persistent data. These security controls address NFR-002 by ensuring authenticated access, role-based authorization, protection of subscriber information, and validation of subscription changes before persistent data is updated.

8. Non-Functional Considerations

8.1 Performance

The Fulfillment Manifest Generator shall be designed to efficiently process a large number of subscription boxes and generate the required shipping information. The manifest-generation component is separated from other fulfillment functionality so that its performance can be optimized independently.

8.2 Maintainability

The layered architecture separates presentation, business logic, and data management.

Changes to the user interface can therefore be made without directly modifying database logic.

Similarly, changes to business rules can be implemented within the relevant business component.

8.3 Modularity

The system is divided into independent components with clearly defined responsibilities.

This allows individual components to be developed, tested, maintained, and modified independently.

8.4 Scalability

The separation of fulfillment and manifest-generation functionality allows the fulfillment subsystem to be optimized independently as the number of subscription boxes increases.

9. Component Diagram

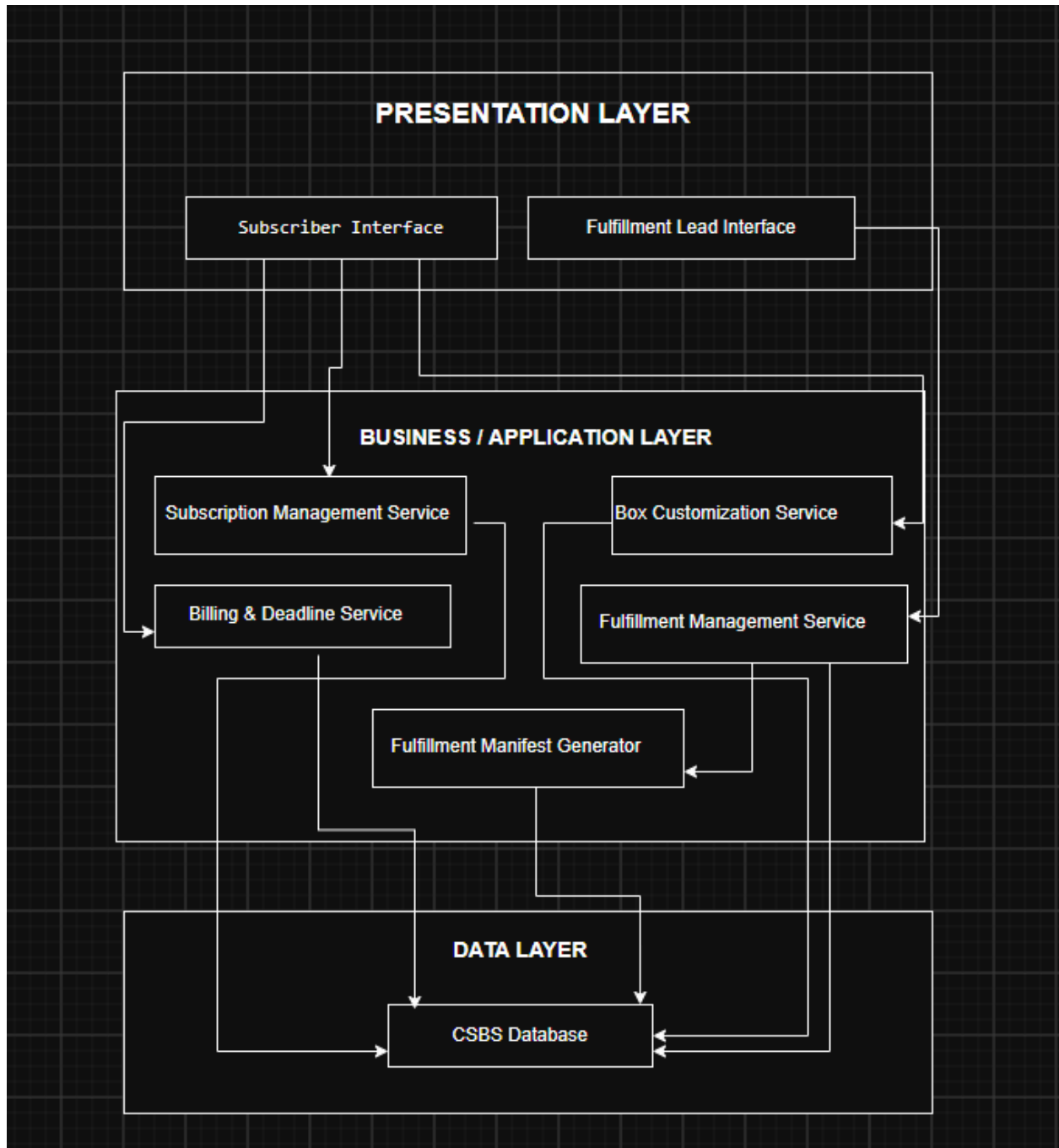


Figure 1: Component Diagram of the Customizable Subscription Box Scheduler.

10. Conclusion

The proposed architecture organizes the Customizable Subscription Box Scheduler into presentation, business/application, and data layers.

The architecture separates user interaction from business logic and persistent data management.

The identified components provide clear responsibilities for subscription management, box customization, deadline validation, fulfillment management, and manifest generation.

The architecture provides traceability between the system requirements and the identified software components while supporting maintainability, modularity, security, and performance.